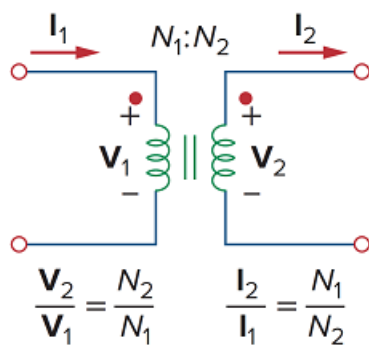


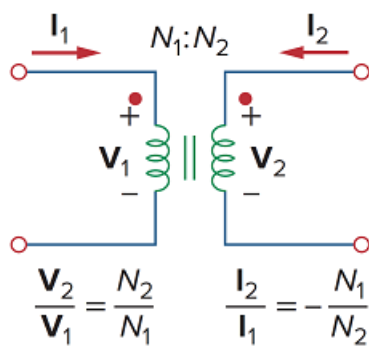
Problem

As done in Fig. 13.32, obtain the relationships between terminal voltages and currents for each of the ideal transformers in Fig. 13.105.

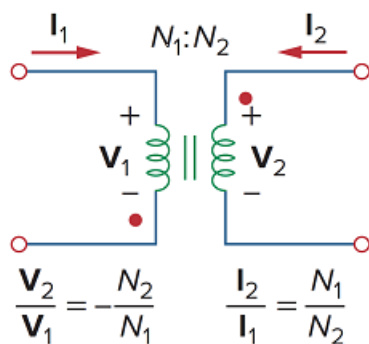
Figure 13.32: Typical circuits illustrating proper voltage polarities and current directions in an ideal transformer.



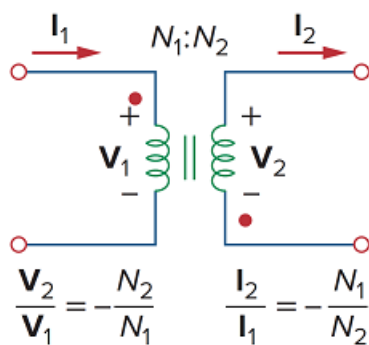
(a)



(b)



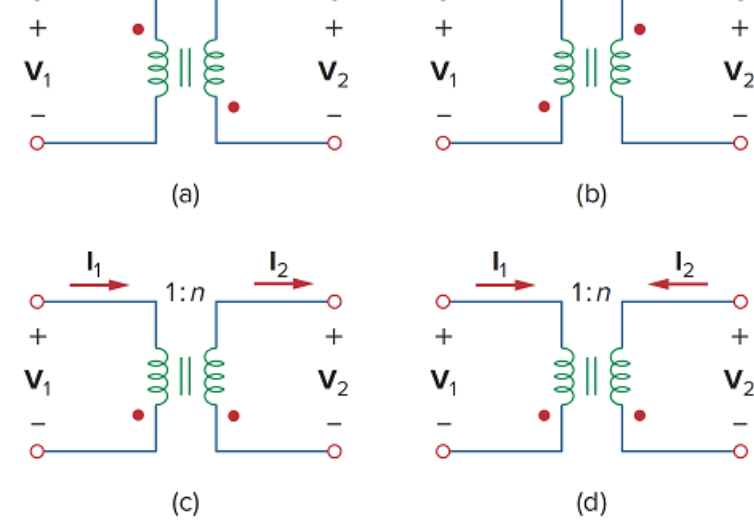
(c)



(d)

Figure 13.105





Step-by-step solution

Step 1 of 4

(a)

Refer to the Figure 13.105 (a) in the textbook.

The voltage V_1 is positive at dotted terminal and voltage V_2 is negative at dotted terminal.

Hence, the transformer ratio is negative.

$$\begin{aligned}\frac{V_2}{V_1} &= -\frac{N_2}{N_1} \\ &= -\frac{n}{1} \\ &= -n\end{aligned}$$

Therefore, the relation between terminal voltage and transformer ratio is $\boxed{\frac{V_2}{V_1} = -n}$.

The current I_1 is entering and current I_2 is leaving the dotted terminal. Hence, the transformer ratio is positive.

$$\begin{aligned}\frac{I_2}{I_1} &= \frac{N_1}{N_2} \\ &= \frac{1}{n}\end{aligned}$$

Therefore, the relation between terminal currents and transformer ratio is $\boxed{\frac{I_2}{I_1} = \frac{1}{n}}$.

Step 2 of 4

(b)

Refer to the Figure 13.105 (b) in the textbook.

The voltage V_1 is negative at dotted terminal and voltage V_2 is positive at dotted terminal.

Hence, the transformer ratio is negative.

$$\begin{aligned}\frac{V_2}{V_1} &= -\frac{N_2}{N_1} \\ &= -\frac{n}{1} \\ &= -n\end{aligned}$$

Therefore, the relation between terminal voltage and transformer ratio is $\boxed{\frac{V_2}{V_1} = -n}$.

The current I_1 and current I_2 are leaving the dotted terminal. Hence, the transformer ratio is negative.

$$\begin{aligned}\frac{I_2}{I_1} &= -\frac{N_1}{N_2} \\ &= -\frac{1}{n}\end{aligned}$$

Therefore, the relation between terminal currents and transformer ratio is $\boxed{\frac{I_2}{I_1} = -\frac{1}{n}}$.

Step 3 of 4

(c)

Refer to the Figure 13.105 (c) in the textbook.

The voltage V_1 is negative at dotted terminal and voltage V_2 is negative at dotted terminal.

Hence, the transformer ratio is positive.

$$\begin{aligned}\frac{V_2}{V_1} &= \frac{N_2}{N_1} \\ &= \frac{n}{1} \\ &= n\end{aligned}$$

Therefore, the relation between terminal voltage and transformer ratio is $\boxed{\frac{V_2}{V_1} = n}$.

The current I_1 is leaving and current I_2 is entering the dotted terminal. Hence, the transformer ratio is positive.

$$\begin{aligned}\frac{I_2}{I_1} &= \frac{N_1}{N_2} \\ &= \frac{1}{n}\end{aligned}$$

Therefore, the relation between terminal currents and transformer ratio is $\boxed{\frac{I_2}{I_1} = \frac{1}{n}}$.

Step 4 of 4

(d)

Refer to the Figure 13.105 (d) in the textbook.

The voltage V_1 is negative at dotted terminal and voltage V_2 is negative at dotted terminal.

Hence, the transformer ratio is positive.

$$\begin{aligned}\frac{V_2}{V_1} &= \frac{N_2}{N_1} \\ &= \frac{n}{1} \\ &= n\end{aligned}$$

Therefore, the relation between terminal voltage and transformer ratio is $\boxed{\frac{V_2}{V_1} = n}$.

The current I_1 is leaving and current I_2 is also leaving the dotted terminal. Hence, the transformer ratio is negative.

$$\begin{aligned}\frac{I_2}{I_1} &= -\frac{N_1}{N_2} \\ &= -\frac{1}{n}\end{aligned}$$

Therefore, the relation between terminal currents and transformer ratio is $\boxed{\frac{I_2}{I_1} = -\frac{1}{n}}$.